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Assignment: Day 01 - .NET Fundamentals & Self-Study

.NET Fundamentals & Self-Study Report

Section 1: .NET Ecosystem Fundamentals

1. .NET Versions

Definition: The .NET platform has undergone a significant architectural evolution since its inception, transitioning from a Windows-only framework into a modern, unified, cross-platform ecosystem.

Historical Evolution

- **.NET Framework (2002):** The original implementation, released by Microsoft in 2002. It is Windows-only, tightly coupled to the operating system, and primarily used for building desktop (WinForms, WPF) and web (ASP.NET Web Forms/MVC) applications on Windows servers.
- **.NET Core (2016):** A ground-up rewrite introduced to address the limitations of .NET Framework. It was designed to be cross-platform (Windows, Linux, macOS), open-source, and modular, allowing developers to deploy applications on non-Windows infrastructure such as Linux-based cloud servers and Docker containers.
- **.NET 5+ (Unified Platform, 2020 onwards):** Starting with .NET 5, Microsoft merged .NET Framework, .NET Core, and Xamarin/Mono into a single, unified platform simply called ".NET" (dropping the word "Core"). This eliminated the confusion of maintaining two parallel frameworks and created one consistent runtime, base class library (BCL), and toolchain for all application types — web, desktop, mobile, cloud, and IoT.

LTS vs. STS Release Cycles

Release Type	Support Duration	Recommended Use Case
LTS (Long-Term Support)	3 years of support	Production systems requiring stability and minimal upgrade churn (e.g., .NET 6, .NET 8)
STS (Standard-Term Support)	18 months of support	Projects that want early access to the latest features and are able to upgrade more frequently (e.g., .NET 7, .NET 9)

Microsoft alternates between LTS and STS releases annually, allowing developers to choose between long-term stability and rapid access to new language and runtime features.

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2. Namespace

Definition: A namespace is a logical container used in C# to organize and group related classes, interfaces, structs, enums, and delegates under a single, meaningful name.

Key Characteristics

- **Prevents Naming Collisions:** Two classes with the identical name (e.g., `Logger`) can coexist in the same project or across referenced libraries as long as they reside in different namespaces (e.g., `Company.Logging.Logger` vs. `ThirdParty.Utils.Logger`).
- **Logical Grouping:** Namespaces group related functionality together, improving code readability and maintainability — similar to how folders organize files on a disk.

- **Hierarchical Structure:** Namespaces can be nested using dot notation (e.g., `System.Collections.Generic`), reflecting increasingly specific categories of functionality.

Translation to Project Structure

In modern .NET projects, namespaces typically mirror the physical folder structure of the solution. For example, a class located at `Services/Payments/PaymentProcessor.cs` would conventionally use the namespace `MyApp.Services.Payments`. This convention keeps the codebase intuitive to navigate, as the namespace path directly reflects where the file lives on disk.

Practical Example:

```
namespace MyApp.Services.Payments
{
    public class PaymentProcessor
    {
        public void ProcessPayment(decimal amount)
        {
            // Business logic here
        }
    }
}

// Usage from another file, after importing the namespace:
using MyApp.Services.Payments;

var processor = new PaymentProcessor();
processor.ProcessPayment(150.00m);
```

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3. .NET Core

Definition: .NET Core was a re-engineered, open-source implementation of the .NET platform, first released in 2016 as a direct response to the limitations of the legacy .NET Framework.

Why It Was Introduced

- **Cross-Platform Support:** Unlike .NET Framework, which was strictly bound to Windows, .NET Core was built to run natively on Windows, Linux, and macOS — enabling developers to deploy applications on cost-effective Linux servers and containerized environments (Docker/Kubernetes).
- **Open-Source Development:** The entire runtime, libraries, and compiler were published on GitHub under an open-source license, allowing community contributions, transparency, and faster innovation.
- **Component-Based Architecture via NuGet:** Instead of shipping a single, monolithic framework, .NET Core adopted a modular design where individual features are distributed as lightweight NuGet packages. This allows applications to include only the components they actually need, reducing deployment size and improving performance.

Market Context

By the mid-2010s, competing technologies such as Node.js, Java (Spring Boot), and Python-based frameworks were gaining significant traction for building modern, high-performance, cloud-native web services — largely because they were free, open-source, and platform-independent. .NET Core was Microsoft's strategic response to this shift, ensuring the .NET ecosystem remained competitive and relevant in the modern cloud and open-source software landscape, rather than losing further market share to these alternatives.

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4. Solution vs. Project

Definition: .NET applications are organized using a two-level hierarchy: the Solution and the Project. Understanding this distinction is essential for structuring real-world, multi-component applications.

Solution (.sln)

- A Solution is a top-level container file that groups one or more related Projects together.
- It does not contain source code itself — it simply defines which projects belong together and how they relate (build order, dependencies).
- Managed and opened through IDEs such as Visual Studio, which use the `.sln` file to load the entire set of related projects at once.

Project (.csproj)

- A Project represents a single, independently buildable unit of software — such as a Web API, a Class Library, a Console Application, or a Mobile App.
- Each project has its own `.csproj` file, which defines its target framework, dependencies (NuGet packages), and build output type (e.g., DLL or executable).
- A single Solution commonly contains multiple Projects that work together, such as separating an application into distinct architectural layers.

Practical Example — Typical Layered Architecture

Solution: ECommerceApp.sln	Project Type
ECommerceApp.Api	ASP.NET Core Web API (presentation layer)
ECommerceApp.Application	Class Library (business logic layer)
ECommerceApp.Domain	Class Library (entities and domain models)
ECommerceApp.Infrastructure	Class Library (data access, external services)
ECommerceApp.Tests	Unit Test Project

> In this structure, the **Solution** (`ECommerceApp.sln`) ties all five independent **Projects** together, allowing Visual Studio to build, debug, and manage inter-project references as a single cohesive application.

Section 2: Bonus (Self-Study Report)

Reducing JIT (Just-In-Time) Compilation Overhead in .NET

Research Question: What is done to reduce the runtime overhead caused by JIT (Just-In-Time) compilation in .NET?

Background

In .NET, source code (C#) is first compiled by the Roslyn compiler into an intermediate, platform-independent format called **Intermediate Language (IL)**. When the application runs, the Common Language Runtime (CLR) uses a **Just-In-Time (JIT) compiler** to translate this IL into native machine code that the processor can execute directly. While this design provides platform independence and runtime optimizations, JIT compilation itself introduces a measurable performance cost, particularly during application startup. The CLR employs several techniques to minimize this overhead.

1. Caching

The JIT compiler does not recompile a method every time it is called. Instead, a method's IL is compiled to native code only on its **first invocation**. The resulting native code is then cached in memory, and every subsequent call to that same method reuses the cached native code directly — avoiding redundant compilation work for the remainder of the application's execution.

2. Pre-Compilation (AOT / NGEN / ReadyToRun)

To eliminate JIT overhead at startup entirely (or significantly reduce it), .NET provides several ahead-of-time compilation technologies that translate IL into native code **before** the application is executed:

- **NGEN (Native Image Generator):** A legacy tool used with .NET Framework that pre-compiles assemblies into native images and stores them in the Native Image Cache on the local machine.
- **ReadyToRun (R2R):** A modern, cross-platform pre-compilation format used in .NET Core/.NET 5+. It embeds native code directly inside the assembly alongside the original IL, allowing the runtime to use the pre-compiled native code immediately at startup while still retaining IL for scenarios requiring re-JITting (e.g., further optimization).
- **Native AOT:** A more aggressive compilation model (introduced in .NET 7+) that compiles the entire application — including the runtime itself — directly into a single, self-contained native executable. This removes the JIT compiler and the need for IL entirely at runtime, resulting in near-instant startup times and a smaller memory footprint, at the cost of some dynamic runtime flexibility (e.g., limited reflection support).

3. Tiered Compilation

Introduced to balance fast startup with long-term execution performance, Tiered Compilation allows the JIT to compile a method twice, using two distinct optimization levels:

- **Tier 0 (Quick JIT):** On first call, the method is compiled rapidly with minimal optimizations. This favors fast startup over raw execution speed, since the compilation itself must be as lightweight as possible.
- **Tier 1 (Optimized JIT):** The CLR monitors method invocation counts in the background. If a method is identified as a **"hot path"** — one that is called frequently — it is automatically re-compiled with full optimizations applied, replacing the Tier 0 version transparently while the application continues running.

> This tiered approach ensures that an application starts up quickly (since most methods only need a fast, unoptimized compilation), while performance-critical, frequently executed code paths still eventually receive the full benefit of aggressive JIT optimizations — achieving the best of both worlds.

Summary Comparison Table

Technique	When It Applies	Primary Benefit
Caching	After a method's first call	Avoids recompiling the same method repeatedly
Pre-Compilation (AOT/R2R/NGEN)	Before the application runs	Eliminates or reduces JIT work at startup
Tiered Compilation	Dynamically, during execution	Fast startup + optimized performance for hot code paths

Conclusion

Together, these three mechanisms — caching, ahead-of-time compilation, and tiered compilation — allow the .NET runtime to strike a practical balance between fast application startup and sustained runtime performance, making JIT overhead largely negligible in well-optimized, modern .NET applications.